API Documentation

AI-Powered Supply Chain Disruption Assistant — REST API (FastAPI)

Overview

Item	Value
Base URL	http://127.0.0.1:8000 (local)
Interactive docs	http://127.0.0.1:8000/docs (Swagger UI) · /redoc · /openapi.json
Format	JSON request and response bodies (Content-Type: application/json); /analyze/stream returns newline-delimited JSON
Start	uvicorn src.main:app --reload
LLM	Claude Sonnet 5.5 via Anthropic's OpenAI-compatible endpoint (LLM_PROVIDER=anthropic); switchable at runtime with POST /llm/provider
Errors	422 validation_error (schema/range violations) or guardrail_violation (unsafe/off-topic input); 400 bad provider switch; 503 deep check without an LLM; 500 unexpected server error

About the samples: every sample response below is a real response from the running service. They were produced with the language model switched off (template mode) so they are reproducible; with Claude switched on, the classification, plan wording and judge scores come from the LLM, while the retrieved cases, predictions, priority and strategy ranking are computed the same way. Long lists are shortened.

Endpoint summary

Method	Endpoint	Purpose
GET	/health	Service status: LLM provider, index and model readiness.
GET	/llm/providers	LLM providers the service can use (whether a key is configured; keys are never returned) and the active one.
POST	/llm/provider	Switch the LLM used by all agents at runtime: anthropic, nvidia, gemini, aicredits, ollama, or 'off' for template mode. 400 if the provider is unknown or has no key.
POST	/classify	Classification agent: extracts disruption type, industry, region, tier, size, backup availability and severity.
POST	/search	Hybrid (or semantic / keyword) search over historical disruptions with metadata filters. auto_filter=true derives type/industry/region filters from the classification agent.
POST	/recommend	Similar cases, strategies ranked on comparable past cases (history + ML what-if + feedback) and an evidence-grounded plan.
POST	/prioritize	Priority score (0-100), level P1-P4 and factor breakdown; ranks a batch. Returns the weights and level cut-offs used.
POST	/predict	Predicted production impact, revenue loss, partial/full recovery (P10-P50-P90) and risk probabilities, plus what-if by response.
POST	/explain	SHAP explanation of a single prediction (optionally with a plain-language narrative from the LLM).
POST	/risk	Disruption-risk probability, resilience score/grade, exposure by disruption type (at the chosen and at the typical severity) and drivers.
POST	/analyze	Full multi-agent workflow (guardrails, classification, investigation, impact, response, recovery, judge, report). Also returns stated assumptions, override and ambiguity notes, and the agent trace.
POST	/analyze/stream	Same workflow and request as /analyze, streamed as newline-delimited JSON (application/x-ndjson): one {"type": "step"} line as each agent finishes, then one {"type": "result"} line with the same payload as /analyze, or {"type": "error"} (422 guardrail / 500). Used by the React UI to show the agents working live.
POST	/analyze/deep-check	Optional DeepEval check of one finished analysis: answer relevancy, faithfulness, analysis correctness and evidence relevance (G-Eval), each with the evaluator's reason. Needs an LLM (about ten calls, roughly a minute); returns 503 with a plain-language reason when no LLM is available.
POST	/judge	LLM-as-Judge evaluation of any analysis/recommendation (5 criteria, pass/fail).
GET	/stats	Dataset analytics by type, industry, region, severity, tier, response and backup.
GET	/insights	Explainable factor analysis (ANOVA/t-tests, severity-adjusted effects), SHAP global importance, key insights, model metrics.
GET	/evaluation	Latest offline evaluation results (retrieval, search, classification, DeepEval, RAGAS, judge, models).
POST	/feedback	Submit a rating, comment, chosen strategy and optionally the real outcome (used for retraining). At least one signal is required.
GET	/feedback/stats	Feedback analytics and the learned strategy multipliers.
POST	/feedback/retrain	Retrain prediction models including user-reported real outcomes.

Error response examples

```
HTTP 422
{
  "error": "validation_error",
  "violations": [
    {
      "field": "filters.industry.0",
      "message": "Input should be 'Electronics', 'Automotive', 'Consumer Goods', 'Pharmaceuticals' or 'Aerospace'"
    }
  ]
}
```

```
HTTP 422
{
  "detail": {
    "error": "guardrail_violation",
    "violations": [
      "Possible prompt-injection attempt detected."
    ]
  }
}
```

```
HTTP 422
{
  "detail": {
    "error": "guardrail_violation",
    "violations": [
      "Query does not appear to be about a supply chain disruption."
    ]
  }
}
```

1. GET /health

Service status: LLM provider, index and model readiness.

Endpoint	HTTP Method
/health	GET

Request format

No request body.

Response format

Key	Type
status	str
llm	dict
models_ready	bool
index_ready	bool

Sample request

```
GET /health
```

Sample response (HTTP 200, lists shortened)

```
{
  "status": "ok",
  "llm": {
    "enabled": false,
    "provider": "anthropic",
    "model": "claude-sonnet-5-5",
    "judge_model": "claude-sonnet-5-5",
    "base_url": "https://api.anthropic.com/v1/"
  },
  "models_ready": true,
  "index_ready": true
}
```

2. GET /llm/providers

LLM providers the service can use (whether a key is configured; keys are never returned) and the active one.

Endpoint	HTTP Method
/llm/providers	GET

Request format

No request body.

Response format

Key	Type
providers	list
active	dict

Sample request

```
GET /llm/providers
```

Sample response (HTTP 200, lists shortened)


```
{
  "providers": [
    {
      "provider": "nvidia",
      "model": "deepseek-ai/deepseek-v4.1-flash",
      "available": true,
      "active": false
    },
    {
      "provider": "gemini",
      "model": "gemini-2.5-flash",
      "available": true,
      "active": false
    },
    "... 3 more"
  ],
  "active": {
    "enabled": false,
    "provider": "anthropic",
    "model": "claude-sonnet-5-5",
    "judge_model": "claude-sonnet-5-5",
    "base_url": "https://api.anthropic.com/v1/"
  }
}
```

3. POST /llm/provider

Switch the LLM used by all agents at runtime: anthropic, nvidia, gemini, aicredits, ollama, or 'off' for template mode. 400 if the provider is unknown or has no key.

Endpoint	HTTP Method
/llm/provider	POST

Request format

Field	Type	Required	Constraints / allowed values
provider	string	yes	

Response format

Key	Type
enabled	bool
provider	str
model	str
judge_model	str
base_url	str

Sample request

```
POST /llm/provider
Content-Type: application/json

{
  "provider": "off"
}
```

Sample response (HTTP 200, lists shortened)

```
{
  "enabled": false,
  "provider": "anthropic",
  "model": "claude-sonnet-5-5",
  "judge_model": "claude-sonnet-5-5",
  "base_url": "https://api.anthropic.com/v1/"
}
```

4. POST /classify

Classification agent: extracts disruption type, industry, region, tier, size, backup availability and severity.

Endpoint	HTTP Method
/classify	POST

Request format

Field	Type	Required	Constraints / allowed values
query	string	yes	min len 3 max len 2000

Response format

Key	Type
disruption_type	str
industry	str
supplier_region	str
supplier_tier	int
supplier_size	NoneType
has_backup_supplier	bool
estimated_severity	int
severity_source	str
confidence	float
reasoning	str
method	str
suggested_filters	dict
guardrail_warnings	list

Sample request

```
POST /classify
Content-Type: application/json

{
  "query": "A typhoon flooded our Tier-1 semiconductor supplier in Taiwan; production halted for weeks, no backup supp
    lier."
}
```

Sample response (HTTP 200, lists shortened)

```
{
  "disruption_type": "Natural Disaster",
  "industry": "Electronics",
  "supplier_region": "Asia-Pacific",
  "supplier_tier": 1,
  "supplier_size": null,
  "has_backup_supplier": false,
  "estimated_severity": 4,
  "severity_source": "cues",
  "confidence": 0.85,
  "reasoning": "Keyword match: type 'Natural Disaster', industry Electronics, region Asia-Pacific; severity 4 inferred
     from scale cue words.",
  "method": "rules",
  "suggested_filters": {
    "disruption_type": "Natural Disaster",
    "industry": "Electronics",
    "supplier_region": "Asia-Pacific"
  },
  "guardrail_warnings": []
}
```

5. POST /search

Hybrid (or semantic / keyword) search over historical disruptions with metadata filters. auto_filter=true derives type/industry/region filters from the classification agent.

Endpoint	HTTP Method
/search	POST

Request format

Field	Type	Required	Constraints / allowed values
query	string	yes	min len 3 max len 2000
top_k	integer	no	>= 1 <= 50 (default 10)
mode	string	no	hybrid, semantic, keyword (default hybrid)
alpha	number	no	>= 0 <= 1 (default 0.6)
filters	object (Filters)	no	disruption_type, industry, supplier_tier, supplier_region, supplier_size, response_type, has_backup_supplier, disruption_severity, min_severity, max_severity
auto_filter	boolean	no	(default False)

Response format

Key	Type
results	list
total_matching_filters	int
mode	str
alpha	float
query	str
filters_applied	dict
classification	NoneType
summary	str
latency_ms	int
guardrail_warnings	list

Sample request

```
POST /search
Content-Type: application/json

{
  "query": "ransomware attack on electronics supplier",
  "top_k": 3,
  "mode": "hybrid",
  "alpha": 0.6,
  "filters": {
    "industry": [
      "Electronics"
    ],
    "min_severity": 3
  }
}
```

Sample response (HTTP 200, lists shortened)


```
{
  "results": [
    {
      "disruption_id": "DIS-006267",
      "disruption_type": "Cyber Attack",
      "industry": "Electronics",
      "supplier_tier": 3,
      "supplier_region": "Asia-Pacific",
      "supplier_size": "Large",
      "has_backup_supplier": true,
      "disruption_severity": 3,
      "production_impact_pct": 12.0,
      "revenue_loss_usd": 1480140.72,
      "response_type": "Alternative Supplier",
      "response_time_days": 5,
      "partial_recovery_days": 17,
      "full_recovery_days": 30,
      "permanent_supplier_change": true,
      "response_effectiveness": 93.0,
      "narrative": "Significant severity 3 Cyber Attack (cyber attack such as ransomware, malware, hacking or IT syste
    ms outage) affecting a Tier-3 large-sized Electronics supplier in Asia-Pacific with a backup supplier available. P
    roductio...",
      "semantic_score": 0.5996,
      "keyword_score": null,
      "semantic_rank": 1,
      "keyword_rank": null,
      "hybrid_score": 0.6
    },
    {
      "disruption_id": "DIS-077298",
      "disruption_type": "Cyber Attack",
      "industry": "Electronics",
      "supplier_tier": 3,
      "supplier_region": "Asia-Pacific",
      "supplier_size": "Large",
      "has_backup_supplier": true,
      "disruption_severity": 3,
      "production_impact_pct": 28.2,
      "revenue_loss_usd": 1397456.85,
      "response_type": "Alternative Supplier",
      "response_time_days": 3,
      "partial_recovery_days": 8,
      "full_recovery_days": 36,
      "permanent_supplier_change": false,
      "response_effectiveness": 89.2,
      "narrative": "Significant severity 3 Cyber Attack (cyber attack such as ransomware, malware, hacking or IT syste
    ms outage) affecting a Tier-3 large-sized Electronics supplier in Asia-Pacific with a backup supplier available. P
    roductio...",
      "semantic_score": 0.5995,
      "keyword_score": null,
      "semantic_rank": 2,
      "keyword_rank": null,
      "hybrid_score": 0.5903
    },
    "... 1 more"
  ],
  "total_matching_filters": 14266,
  "mode": "hybrid",
  "alpha": 0.6,
  "query": "ransomware attack on electronics supplier",
  "filters_applied": {
    "industry": [
      "Electronics"
    ],
    "min_severity": 3
  },
  "classification": null,
  "summary": "3 similar historical disruptions found (most common: Cyber Attack). Responses used in these 3 cases: Alt
    ernative Supplier (3). Median full recovery 36 days, median production impact 28.2%, 67% ended with a permanent su
    pp...",
  "latency_ms": 296,
  "guardrail_warnings": []
}
```

6. POST /recommend

Similar cases, strategies ranked on comparable past cases (history + ML what-if + feedback) and an evidence-grounded plan.

Endpoint	HTTP Method
/recommend	POST

Request format

Field	Type	Required	Constraints / allowed values
query	string	yes	min len 3 max len 2000
profile	object (ProfileOverrides)	no	disruption_type, industry, supplier_tier, supplier_region, supplier_size, has_backup_supplier, disruption_severity
top_k	integer	no	>= 3 <= 50 (default 10)
filters	object (Filters)	no	disruption_type, industry, supplier_tier, supplier_region, supplier_size, response_type, has_backup_supplier, disruption_severity, min_severity, max_severity

Response format

Key	Type
classification	dict
profile	dict
assumptions	list
historical_summary	str
similar_cases	list
strategy_ranking	list
strategy_basis	dict
plan	dict
guardrail_warnings	list

Sample request

```
POST /recommend
Content-Type: application/json

{
  "query": "Dock workers strike at the port is delaying pharma ingredient shipments from India",
  "top_k": 8
}
```

Sample response (HTTP 200, lists shortened)


```
{
  "classification": {
    "disruption_type": "Port Congestion",
    "industry": "Pharmaceuticals",
    "supplier_region": "Asia-Pacific",
    "supplier_tier": null,
    "supplier_size": null,
    "has_backup_supplier": null,
    "estimated_severity": 3,
    "severity_source": "default",
    "confidence": 0.7,
    "reasoning": "Keyword match: type 'Port Congestion', industry Pharmaceuticals, region Asia-Pacific; severity 3 ass
    umed: nothing in the text says how big or how long.",
    "method": "rules",
    "suggested_filters": {
      "disruption_type": "Port Congestion",
      "industry": "Pharmaceuticals",
      "supplier_region": "Asia-Pacific"
    }
  },
  "profile": {
    "disruption_type": "Port Congestion",
    "industry": "Pharmaceuticals",
    "supplier_tier": 3,
    "supplier_region": "Asia-Pacific",
    "supplier_size": "Small",
    "has_backup_supplier": false,
    "disruption_severity": 3
  },
  "assumptions": [
    "supplier_tier not stated; assumed '3' (most common among similar cases)",
    "supplier_size not stated; assumed 'Small' (most common among similar cases)",
    "... 2 more"
  ],
  "historical_summary": "8 similar historical disruptions found (most common: Port Congestion). Responses used in thes
    e 8 cases: Customer Delay (8). Median full recovery 99 days, median production impact 26.3%, 25% ended with a perm
    anent supplie...",
  "similar_cases": [
    {
      "disruption_id": "DIS-009197",
      "disruption_type": "Port Congestion",
      "industry": "Pharmaceuticals",
      "supplier_tier": 3,
      "supplier_region": "Asia-Pacific",
      "supplier_size": "Small",
      "has_backup_supplier": false,
      "disruption_severity": 3,
      "production_impact_pct": 40.3,
      "revenue_loss_usd": 321604.58,
      "response_type": "Customer Delay",
      "response_time_days": 4,
      "partial_recovery_days": 28,
      "full_recovery_days": 137,
      "permanent_supplier_change": false,
      "response_effectiveness": 17.5,
      "narrative": "Significant severity 3 Port Congestion (port congestion, shipping backlog, container and vessel de
    lays, logistics bottleneck) affecting a Tier-3 small-sized Pharmaceuticals supplier in Asia-Pacific with no backup
     supplie...",
      "semantic_score": 0.4297,
      "keyword_score": 1.9768,
      "semantic_rank": 11,
      "keyword_rank": 27,
      "hybrid_score": 0.796
    },
    {
      "disruption_id": "DIS-005544",
      "disruption_type": "Port Congestion",
      "industry": "Pharmaceuticals",
      "supplier_tier": 3,
      "supplier_region": "Asia-Pacific",
      "supplier_size": "Medium",
      "has_backup_supplier": false,
      "disruption_severity": 3,
      "production_impact_pct": 16.5,
```

```
      "revenue_loss_usd": 433911.2,
... (112 more lines)
```

7. POST /prioritize

Priority score (0-100), level P1-P4 and factor breakdown; ranks a batch. Returns the weights and level cut-offs used.

Endpoint	HTTP Method
/prioritize	POST

Request format

Field	Type	Required	Constraints / allowed values
items	array[PrioritizeItem]	yes	each item: id, disruption_severity, production_impact_pct, revenue_loss_usd, supplier_tier, has_backup_supplier, description min items 1 max items 500

Response format

Key	Type
ranked	list
weights	dict
levels	list

Sample request

```
POST /prioritize
Content-Type: application/json

{
  "items": [
    {
      "id": "Typhoon - Taiwan chips",
      "disruption_severity": 4,
      "production_impact_pct": 60,
      "revenue_loss_usd": 8000000,
      "supplier_tier": 1,
      "has_backup_supplier": false
    },
    {
      "id": "Port backlog - Rotterdam",
      "disruption_severity": 2,
      "production_impact_pct": 18,
      "revenue_loss_usd": 400000,
      "supplier_tier": 2,
      "has_backup_supplier": true
    }
  ]
}
```

Sample response (HTTP 200, lists shortened)


```
{
  "ranked": [
    {
      "id": "Typhoon - Taiwan chips",
      "disruption_severity": 4,
      "production_impact_pct": 60.0,
      "revenue_loss_usd": 8000000.0,
      "supplier_tier": 1,
      "has_backup_supplier": false,
      "description": null,
      "priority_score": 77.5,
      "priority_level": "P1",
      "priority_label": "Critical",
      "recommended_action": "Escalate immediately; activate crisis team and backup sourcing within 24h.",
      "factor_contributions": {
        "severity": 26.25,
        "production_impact": 15.0,
        "financial_loss": 21.24,
        "supplier_tier": 10.0,
        "no_backup": 5.0
      },
      "normalized_factors": {
        "severity": 0.75,
        "production_impact": 0.6,
        "financial_loss": 0.849,
        "supplier_tier": 1.0,
        "no_backup": 1.0
      },
      "rank": 1
    },
    {
      "id": "Port backlog - Rotterdam",
      "disruption_severity": 2,
      "production_impact_pct": 18.0,
      "revenue_loss_usd": 400000.0,
      "supplier_tier": 2,
      "has_backup_supplier": true,
      "description": null,
      "priority_score": 32.1,
      "priority_level": "P3",
      "priority_label": "Medium",
      "recommended_action": "Plan mitigation this week; monitor supplier recovery closely.",
      "factor_contributions": {
        "severity": 8.75,
        "production_impact": 4.5,
        "financial_loss": 11.39,
        "supplier_tier": 7.5,
        "no_backup": 0.0
      },
      "normalized_factors": {
        "severity": 0.25,
        "production_impact": 0.18,
        "financial_loss": 0.455,
        "supplier_tier": 0.75,
        "no_backup": 0.0
      },
      "rank": 2
    }
  ],
  "weights": {
    "severity": 0.35,
    "production_impact": 0.25,
    "financial_loss": 0.25,
    "supplier_tier": 0.1,
    "no_backup": 0.05
  },
  "levels": [
    {
      "min_score": 70,
      "level": "P1",
... (10 more lines)
```

8. POST /predict

Predicted production impact, revenue loss, partial/full recovery (P10-P50-P90) and risk probabilities, plus what-if by response.

Endpoint	HTTP Method
/predict	POST

Request format

Field	Type	Required	Constraints / allowed values
disruption_type	string	yes	Port Congestion, Geopolitical, Labor Strike, Natural Disaster, Cyber Attack, Factory Incident
industry	string	yes	Electronics, Automotive, Consumer Goods, Pharmaceuticals, Aerospace
supplier_tier	integer	yes	1, 2, 3, 4
supplier_region	string	yes	Asia-Pacific, Europe, North America, South America, Africa/Middle East
supplier_size	string	no	Small, Medium, Large (default Medium)
has_backup_supplier	boolean	no	(default False)
disruption_severity	integer	yes	>= 1 <= 5
response_type	string	no	Alternative Supplier, Production Reroute, Inventory Buffer, Customer Delay, Combined Strategy

Response format

Key	Type
profile	dict
predictions	dict
what_if_by_response	list

Sample request

```
POST /predict
Content-Type: application/json

{
  "disruption_type": "Cyber Attack",
  "industry": "Automotive",
  "supplier_tier": 2,
  "supplier_region": "Europe",
  "supplier_size": "Large",
  "has_backup_supplier": true,
  "disruption_severity": 3
}
```

Sample response (HTTP 200, lists shortened)


```
{
  "profile": {
    "disruption_type": "Cyber Attack",
    "industry": "Automotive",
    "supplier_tier": 2,
    "supplier_region": "Europe",
    "supplier_size": "Large",
    "has_backup_supplier": true,
    "disruption_severity": 3,
    "response_type": null
  },
  "predictions": {
    "production_impact_pct": {
      "predicted": 38.1,
      "low_p10": 25.39,
      "high_p90": 50.87
    },
    "revenue_loss_usd": {
      "predicted": 5150702.78,
      "low_p10": 1853359.66,
      "high_p90": 15997731.15
    },
    "full_recovery_days": {
      "predicted": 72.23,
      "low_p10": 28.63,
      "high_p90": 124.74
    },
    "partial_recovery_days": {
      "predicted": 26.36,
      "low_p10": 6.07,
      "high_p90": 52.23
    },
    "high_impact_probability": 0.5931,
    "supplier_change_probability": 0.1657
  },
  "what_if_by_response": [
    {
      "response_type": "Alternative Supplier",
      "predicted_full_recovery_days": 72.23,
      "predicted_partial_recovery_days": 26.36,
      "predicted_revenue_loss_usd": 5150703.0,
      "predicted_production_impact_pct": 38.1,
      "supplier_change_probability": 0.1657
    },
    {
      "response_type": "Production Reroute",
      "predicted_full_recovery_days": 84.57,
      "predicted_partial_recovery_days": 32.12,
      "predicted_revenue_loss_usd": 5230584.0,
      "predicted_production_impact_pct": 38.09,
      "supplier_change_probability": 0.1637
    },
    "... 3 more"
  ]
}
```

9. POST /explain

SHAP explanation of a single prediction (optionally with a plain-language narrative from the LLM).

Endpoint	HTTP Method
/explain	POST

Request format

Field	Type	Required	Constraints / allowed values
profile	object (DisruptionProfile)	yes	disruption_type, industry, supplier_tier, supplier_region, supplier_size, has_backup_supplier, disruption_severity, response_type
target	string	no	production_impact_pct, revenue_loss_usd, full_recovery_days, partial_recovery_days, high_impact, permanent_supplier_change (default full_recovery_days)
narrate	boolean	no	(default False)

Response format

Key	Type
target	str
unit	str
baseline	float
prediction	float
contributions	list
summary	str

Sample request

```
POST /explain
Content-Type: application/json

{
  "profile": {
    "disruption_type": "Natural Disaster",
    "industry": "Electronics",
    "supplier_tier": 1,
    "supplier_region": "Asia-Pacific",
    "supplier_size": "Medium",
    "has_backup_supplier": false,
    "disruption_severity": 4
  },
  "target": "full_recovery_days"
}
```

Sample response (HTTP 200, lists shortened)


```
{
  "target": "full_recovery_days",
  "unit": "days",
  "baseline": 75.853,
  "prediction": 132.035,
  "contributions": [
    {
      "feature": "disruption_severity",
      "label": "Severity",
      "value": "4",
      "contribution": 45.72,
      "direction": "increases"
    },
    {
      "feature": "has_backup_supplier",
      "label": "Backup supplier",
      "value": "No",
      "contribution": 16.654,
      "direction": "increases"
    },
    "... 6 more"
  ],
  "summary": "Main drivers of full recovery days: Severity = 4 increases it (+45.72 days); Backup supplier = No increa
    ses it (+16.65 days); Response strategy = Alternative Supplier decreases it (-6.22 days)."
}
```

10. POST /risk

Disruption-risk probability, resilience score/grade, exposure by disruption type (at the chosen and at the typical severity) and drivers.

Endpoint	HTTP Method
/risk	POST

Request format

Field	Type	Required	Constraints / allowed values
industry	string	yes	Electronics, Automotive, Consumer Goods, Pharmaceuticals, Aerospace
supplier_region	string	yes	Asia-Pacific, Europe, North America, South America, Africa/Middle East
supplier_tier	integer	no	1, 2, 3, 4 (default 1)
supplier_size	string	no	Small, Medium, Large (default Medium)
has_backup_supplier	boolean	no	(default False)
assumed_severity	integer	no	>= 1 <= 5 (default 3)

Response format

Key	Type
profile	dict
assumed_severity	int
disruption_risk_probability	float
risk_band	str
resilience_score	float
resilience_grade	str
exposure_by_type	list
top_risk_drivers	list
recommendations	list

Sample request

```
POST /risk
Content-Type: application/json

{
  "industry": "Aerospace",
  "supplier_region": "Africa/Middle East",
  "supplier_tier": 3,
  "supplier_size": "Small",
  "has_backup_supplier": false,
  "assumed_severity": 3
}
```

Sample response (HTTP 200, lists shortened)


```
{
  "profile": {
    "industry": "Aerospace",
    "supplier_region": "Africa/Middle East",
    "supplier_tier": 3,
    "supplier_size": "Small",
    "has_backup_supplier": false
  },
  "assumed_severity": 3,
  "disruption_risk_probability": 0.1469,
  "risk_band": "Low",
  "resilience_score": 61.3,
  "resilience_grade": "B",
  "exposure_by_type": [
    {
      "disruption_type": "Labor Strike",
      "historical_frequency": 0.143,
      "typical_severity": 1.99,
      "vulnerability_at_typical_severity": 0.226,
      "high_impact_probability": 0.1689,
      "predicted_full_recovery_days": 105.56,
      "predicted_production_impact_pct": 32.46,
      "predicted_revenue_loss_usd": 981499.2,
      "vulnerability": 0.406
    },
    {
      "disruption_type": "Factory Incident",
      "historical_frequency": 0.117,
      "typical_severity": 2.26,
      "vulnerability_at_typical_severity": 0.264,
      "high_impact_probability": 0.1738,
      "predicted_full_recovery_days": 109.44,
      "predicted_production_impact_pct": 30.06,
      "predicted_revenue_loss_usd": 864369.19,
      "vulnerability": 0.404
    },
    "... 4 more"
  ],
  "top_risk_drivers": [
    {
      "feature": "disruption_severity",
      "label": "Severity",
      "value": "3",
      "contribution": 1.439,
      "direction": "increases"
    },
    {
      "feature": "industry",
      "label": "Industry",
      "value": "Aerospace",
      "contribution": 1.195,
      "direction": "increases"
    },
    "... 3 more"
  ],
  "recommendations": [
    "Qualify a backup supplier: backup availability is associated with lower impact and faster recovery.",
    "Map sub-tier suppliers: deep-tier disruptions are detected late; add visibility/early warning.",
    "... 1 more"
  ]
}
```

11. POST /analyze

Full multi-agent workflow (guardrails, classification, investigation, impact, response, recovery, judge, report). Also returns stated assumptions, override and ambiguity notes, and the agent trace.

Endpoint	HTTP Method
/analyze	POST

Request format

Field	Type	Required	Constraints / allowed values
query	string	yes	min len 3 max len 2000
profile	object (ProfileOverrides)	no	disruption_type, industry, supplier_tier, supplier_region, supplier_size, has_backup_supplier, disruption_severity
production_impact_pct	number	no	>= 0 <= 100
revenue_loss_usd	number	no	>= 0 <= 100000000000.0
filters	object (Filters)	no	disruption_type, industry, supplier_tier, supplier_region, supplier_size, response_type, has_backup_supplier, disruption_severity, min_severity, max_severity
top_k	integer	no	>= 3 <= 30 (default 10)

Response format

Key	Type
query	str
executive_summary	str
classification	dict
profile	dict
assumptions	list
override_notes	list
ambiguity_notes	list
investigation	dict
impact_assessment	dict
response	dict
recovery	dict
judgement	dict
revisions	int
trace	list
guardrail_warnings	list
total_duration_ms	int

Sample request

```
POST /analyze
Content-Type: application/json

{
  "query": "Ransomware has locked the ERP of our Tier-2 auto parts supplier in Germany; shipments stopped."
}
```

Sample response (HTTP 200, lists shortened)


```
{
  "query": "Ransomware has locked the ERP of our Tier-2 auto parts supplier in Germany; shipments stopped.",
  "executive_summary": "P2 (High) Cyber Attack disruption affecting a Tier-2 Automotive supplier in Europe. Expected p
    roduction impact 52% and revenue loss $6,674,257. Recommended response: Inventory Buffer, with partial recovery in
     ~34 days an...",
  "classification": {
    "disruption_type": "Cyber Attack",
    "industry": "Automotive",
    "supplier_region": "Europe",
    "supplier_tier": 2,
    "supplier_size": null,
    "has_backup_supplier": null,
    "estimated_severity": 4,
    "severity_source": "cues",
    "confidence": 0.85,
    "reasoning": "Keyword match: type 'Cyber Attack', industry Automotive, region Europe; severity 4 inferred from sca
    le cue words.",
    "method": "rules",
    "suggested_filters": {
      "disruption_type": "Cyber Attack",
      "industry": "Automotive",
      "supplier_region": "Europe"
    }
  },
  "profile": {
    "disruption_type": "Cyber Attack",
    "industry": "Automotive",
    "supplier_tier": 2,
    "supplier_region": "Europe",
    "supplier_size": "Large",
    "has_backup_supplier": true,
    "disruption_severity": 4
  },
  "assumptions": [
    "supplier_size not stated; assumed 'Large' (most common among similar cases)",
    "has_backup_supplier not stated; assumed 'True' (most common among similar cases)"
  ],
  "override_notes": [],
  "ambiguity_notes": [],
  "investigation": {
    "historical_summary": "10 similar historical disruptions found (most common: Cyber Attack). Responses used in thes
    e 10 cases: Alternative Supplier (6), Inventory Buffer (4). Median full recovery 78 days, median production impact
     38.5%, 30% end...",
    "filters_applied": {
      "disruption_type": "Cyber Attack",
      "industry": "Automotive",
      "supplier_region": "Europe",
      "min_severity": 3,
      "max_severity": 5
    },
    "relaxations": [],
    "total_matching_filters": 604,
    "similar_cases": [
      {
        "disruption_id": "DIS-068423",
        "disruption_type": "Cyber Attack",
        "industry": "Automotive",
        "supplier_tier": 2,
        "supplier_region": "Europe",
        "supplier_size": "Medium",
        "has_backup_supplier": true,
        "disruption_severity": 3,
        "production_impact_pct": 40.3,
        "revenue_loss_usd": 2296299.79,
        "response_type": "Alternative Supplier",
        "response_time_days": 2,
        "partial_recovery_days": 23,
        "full_recovery_days": 86,
        "permanent_supplier_change": false,
        "response_effectiveness": 44.9,
        "narrative": "Significant severity 3 Cyber Attack (cyber attack such as ransomware, malware, hacking or IT sys
    tems outage) affecting a Tier-2 medium-sized Automotive supplier in Europe with a backup supplier available. Produ
    ction drop...",
        "semantic_score": 0.4409,
```

```
        "keyword_score": 3.8748,
        "semantic_rank": 15,
        "keyword_rank": 4,
... (342 more lines)
```

12. POST /analyze/stream

Same workflow and request as /analyze, streamed as newline-delimited JSON (application/x-ndjson): one {"type": "step"} line as each agent finishes, then one {"type": "result"} line with the same payload as /analyze, or {"type": "error"} (422 guardrail / 500). Used by the React UI to show the agents working live.

Endpoint	HTTP Method
/analyze/stream	POST

Request format

Field	Type	Required	Constraints / allowed values
query	string	yes	min len 3 max len 2000
profile	object (ProfileOverrides)	no	disruption_type, industry, supplier_tier, supplier_region, supplier_size, has_backup_supplier, disruption_severity
production_impact_pct	number	no	>= 0 <= 100
revenue_loss_usd	number	no	>= 0 <= 100000000000.0
filters	object (Filters)	no	disruption_type, industry, supplier_tier, supplier_region, supplier_size, response_type, has_backup_supplier, disruption_severity, min_severity, max_severity
top_k	integer	no	>= 3 <= 30 (default 10)

Response format

Line type	Fields	When
step	type, node, step {agent, status, duration_ms, summary}	each time an agent finishes
result	type, data (same object as the /analyze response)	once, at the end
error	type, status (422 or 500), violations or message	guardrail block or failure

Sample request

```
POST /analyze/stream
Content-Type: application/json

{
  "query": "Ransomware has locked the ERP of our Tier-2 auto parts supplier in Germany; shipments stopped."
}
```

Sample response (HTTP 200, 9 lines; the result line is shortened)


```
{"type": "step", "node": "guardrail", "step": {"agent": "Guardrails", "status": "ok", "duration_ms": 4, "summary": "In
    put accepted"}}
{"type": "step", "node": "classify", "step": {"agent": "Classification Agent", "status": "ok", "duration_ms": 2, "summ
    ary": "Cyber Attack | severity 4 | Automotive | Europe (conf 0.85, rules)"}}
{"type": "step", "node": "investigate", "step": {"agent": "Investigation Agent", "status": "ok", "duration_ms": 314, "
    summary": "10 similar cases retrieved from 604 matching records; filters {'disruption_type': 'Cyber Attack', 'indu
    stry': 'Automotive', 'supplier_region': 'Europe', 'min_severity': 3, 'max_severity': 5}"}}
{"type": "step", "node": "impact", "step": {"agent": "Impact Assessment Agent", "status": "ok", "duration_ms": 4107, "
    summary": "Impact 51.7%, loss $6,674,257, full recovery 99d -> P2 High (67.3)"}}
{"type": "step", "node": "response", "step": {"agent": "Response Planning Agent", "status": "ok", "duration_ms": 264,
    "summary": "Recommended Inventory Buffer (template); 5 strategies ranked"}}
{"type": "step", "node": "recovery", "step": {"agent": "Recovery Analysis Agent", "status": "ok", "duration_ms": 5666,
     "summary": "Full recovery ~99d; supplier change p=0.35; resilience 29.0"}}
{"type": "step", "node": "judge", "step": {"agent": "Judge Agent (LLM-as-Judge)", "status": "ok", "duration_ms": 0, "s
    ummary": "Overall 4.6/5 -> PASS (heuristic)"}}
{"type": "step", "node": "report", "step": {"agent": "Report", "status": "ok", "duration_ms": 0, "summary": "Final rep
    ort compiled"}}
{"type": "result", "data": {"query": "Ransomware has locked the ERP of our Tier-2 auto parts supplier in Germany; ship
    ments stopped.", "executive_summary": "P2 (High) Cyber Attack disruption affecting a Tier-2 Automotive supplier in
     Europe. Expected production impact 52% and revenue loss $6,674,257. Recommended response: Inventory Buffer, with
    partial recovery in ~34 days an...", "classification": {"disruption_type": "Cyber Attack", "industry": "Automotive
    ", "supplier_region": "Europe", "supplier_tier": 2, "supplier_size": null, "has_backup_supplier": null, "estimated
    _severity": 4, "severity_source": "cues", "confidence": 0.85, "reasoning": "Keyword match: type 'Cyber Attack', in
    dustry Automotive, region Europe; severity 4 inferred from scale cue words.", "method": "rules", "suggested_filter
    s": {"disruption_type": "Cyber Attack", "industry": "Automotive", "supplier_region": "Europe"}}, "profile": {"disr
    uption_type": "Cyber Attack", "industry": "Automotive", "supplier_tier": 2, "supplier_region": "Europe", "supplier
    _size": "Large", "has_backup_supplier": true, "disruption_severity": 4}, "...": "same keys as /analyze"}}
```

13. POST /analyze/deep-check

Optional DeepEval check of one finished analysis: answer relevancy, faithfulness, analysis correctness and evidence relevance (G-Eval), each with the evaluator's reason. Needs an LLM (about ten calls, roughly a minute); returns 503 with a plain-language reason when no LLM is available.

Endpoint	HTTP Method
/analyze/deep-check	POST

Request format

Field	Type	Required	Constraints / allowed values
query	string	yes	min len 3 max len 2000
plan	object	yes	The response plan to check, including its fact_sheet (the evidence it was given)

Response format

Success (200):

Key	Type	Meaning
framework	str	"DeepEval"
judge_model	str	evaluator model
n_contexts	int	evidence lines the plan was checked against (its fact sheet)
seconds	float	time taken
scores	dict	per metric: {score 0-1, passed, reason}; null if that metric failed
errors	dict	per failed metric: short reason

Sample request

```
POST /analyze/deep-check
Content-Type: application/json

{
  "query": "Dock workers strike at the port is delaying pharma ingredient shipments from India",
  "plan": {
    "recommended_strategy": "Inventory Buffer",
    "rationale": "Inventory Buffer has the best overall score among the options: ~105 days to full recovery (model est
    imate), permanent supplier-change risk 23%, used in 116 of 420 comparable past cases.",
    "immediate_actions": [
      "Confirm scope: affected parts, inventory on hand and customer orders at risk",
      "Draw down safety stock and allocate inventory to priority customers",
      "... 1 more"
    ],
    "recovery_plan": [
      "Target partial recovery within ~35 days",
      "Target full recovery within ~105 days",
      "... 1 more"
    ],
    "fallback_strategy": "Production Reroute",
    "risks": [
      "Permanent supplier change risk ~23%",
      "Recovery may run longer if severity escalates and there is no qualified backup supplier to fall back on"
    ],
    "cited_cases": [],
    "generated_by": "template",
    "fact_sheet": "DISRUPTION PROFILE - Port Congestion, severity 3, Tier-3 Pharmaceuticals supplier in Asia-Pacific,
    no backup supplier.\nPAST OUTCOMES IN 420 COMPARABLE CASES (context only; recovery times for this disruption are t
    he model..."
  }
}
```

Sample response (HTTP 503: these samples were generated with no LLM, so the service explains why it cannot run the check)

```
{
  "detail": "The deep check needs a language model, and none is available. Pick one in the model selector."
}
```

14. POST /judge

LLM-as-Judge evaluation of any analysis/recommendation (5 criteria, pass/fail).

Endpoint	HTTP Method
/judge	POST

Request format

Field	Type	Required	Constraints / allowed values
query	string	yes	min len 3 max len 2000
answer	object	yes	The analysis/recommendation to evaluate
evidence	object	no	Retrieved evidence used to produce the answer

Response format

Key	Type
scores	dict
reasons	dict
overall_feedback	str
judge	str
fallback_reason	str
overall_score	float
passed	bool
threshold	float

Sample request

```
POST /judge
Content-Type: application/json

{
  "query": "Port congestion delaying electronics components from Vietnam",
  "answer": {
    "response_plan": {
      "recommended_strategy": "Inventory Buffer",
      "rationale": "Similar cases DIS-000003 recovered fastest.",
      "immediate_actions": [
        "Draw down safety stock",
        "Expedite air freight",
        "Notify customers"
      ],
      "recovery_plan": [
        "Partial recovery in 10 days",
        "Full recovery in 30 days"
      ],
      "risks": [
        "Stock-outs"
      ],
      "cited_cases": [
        "DIS-000003"
      ]
    }
  },
  "evidence": {
    "case_ids": [
      "DIS-000003"
    ],
    "classification": {
      "disruption_type": "Port Congestion"
    }
  }
}
```

Sample response (HTTP 200, lists shortened)


```
{
  "scores": {
    "faithfulness": 5,
    "relevance": 4,
    "actionability": 4,
    "completeness": 3,
    "safety": 5
  },
  "reasons": {
    "faithfulness": "Cites 1 retrieved case(s)",
    "relevance": "Strategy and classification align with the described disruption.",
    "actionability": "3 immediate actions; timeline present.",
    "completeness": "3/5 sections present (impact, priority, strategy, recovery plan, risks).",
    "safety": "No unsafe recommendations detected by rule checks."
  },
  "overall_feedback": "Checklist evaluation. No language model is switched on.",
  "judge": "heuristic",
  "fallback_reason": "no language model switched on",
  "overall_score": 4.2,
  "passed": true,
  "threshold": 3.5
}
```

15. GET /stats

Dataset analytics by type, industry, region, severity, tier, response and backup.

Endpoint	HTTP Method
/stats	GET

Request format

No request body.

Response format

Key	Type
total_records	int
totals	dict
by_disruption_type	dict
by_industry	dict
by_region	dict
by_severity	dict
by_tier	dict
by_response_type	dict
by_backup_supplier	dict
recovery_days_by_type_and_response	dict

Sample request

```
GET /stats
```

Sample response (HTTP 200, lists shortened)


```
{
  "total_records": 100000,
  "totals": {
    "total_revenue_loss_usd": 252281185565.0,
    "avg_production_impact_pct": 31.4,
    "median_full_recovery_days": 61.0,
    "backup_supplier_rate": 0.556,
    "permanent_supplier_change_rate": 0.239
  },
  "by_disruption_type": {
    "Cyber Attack": {
      "count": 14887,
      "avg_production_impact_pct": 41.3,
      "avg_revenue_loss_usd": 3365434.0,
      "avg_full_recovery_days": 99.6,
      "supplier_change_rate": 0.294
    },
    "Factory Incident": {
      "count": 11878,
      "avg_production_impact_pct": 29.6,
      "avg_revenue_loss_usd": 2359573.0,
      "avg_full_recovery_days": 71.3,
      "supplier_change_rate": 0.224
    },
    "Geopolitical": {
      "count": 18084,
      "avg_production_impact_pct": 33.9,
      "avg_revenue_loss_usd": 2691691.0,
      "avg_full_recovery_days": 81.2,
      "supplier_change_rate": 0.252
    },
    "Labor Strike": {
      "count": 15017,
      "avg_production_impact_pct": 26.0,
      "avg_revenue_loss_usd": 2033352.0,
      "avg_full_recovery_days": 62.8,
      "supplier_change_rate": 0.212
    },
    "Natural Disaster": {
      "count": 15016,
      "avg_production_impact_pct": 37.3,
      "avg_revenue_loss_usd": 3039002.0,
      "avg_full_recovery_days": 90.6,
      "supplier_change_rate": 0.268
    },
    "Port Congestion": {
      "count": 25118,
      "avg_production_impact_pct": 24.3,
      "avg_revenue_loss_usd": 1963051.0,
      "avg_full_recovery_days": 59.1,
      "supplier_change_rate": 0.204
    }
  },
  "by_industry": {
    "Aerospace": {
      "count": 10049,
      "avg_production_impact_pct": 31.4,
      "avg_revenue_loss_usd": 6067862.0,
      "avg_full_recovery_days": 75.6,
      "supplier_change_rate": 0.228
    },
    "Automotive": {
      "count": 24934,
      "avg_production_impact_pct": 31.4,
      "avg_revenue_loss_usd": 2432135.0,
      "avg_full_recovery_days": 75.6,
      "supplier_change_rate": 0.239
    },
    "Consumer Goods": {
      "count": 20158,
... (221 more lines)
```

16. GET /insights

Explainable factor analysis (ANOVA/t-tests, severity-adjusted effects), SHAP global importance, key insights, model metrics.

Endpoint	HTTP Method
/insights	GET

Request format

No request body.

Response format

Key	Type
key_insights	list
factor_analysis	dict
global_importance	dict
model_metrics	dict

Sample request

```
GET /insights
```

Sample response (HTTP 200, lists shortened)


```
{
  "key_insights": [
    "Severity is a key driver of production impact % (explains 57.3% of variance, p<0.001): severity 5 averages 62.2%
    vs severity 1 13.3%.",
    "Supplier tier is a key driver of production impact % (explains 9.6% of variance, p<0.001): 1 averages 37.4% vs 4
    21.0%.",
    "... 5 more"
  ],
  "factor_analysis": {
    "disruption_type": {
      "label": "Disruption type",
      "groups": {
        "Cyber Attack": {
          "count": 14887,
          "production_impact_pct": 41.31,
          "revenue_loss_usd": 3365433.78,
          "full_recovery_days": 99.62,
          "permanent_supplier_change_rate": 0.294
        },
        "Factory Incident": {
          "count": 11878,
          "production_impact_pct": 29.58,
          "revenue_loss_usd": 2359573.42,
          "full_recovery_days": 71.27,
          "permanent_supplier_change_rate": 0.224
        },
        "Geopolitical": {
          "count": 18084,
          "production_impact_pct": 33.86,
          "revenue_loss_usd": 2691690.6,
          "full_recovery_days": 81.17,
          "permanent_supplier_change_rate": 0.252
        },
        "Labor Strike": {
          "count": 15017,
          "production_impact_pct": 26.0,
          "revenue_loss_usd": 2033352.27,
          "full_recovery_days": 62.77,
          "permanent_supplier_change_rate": 0.212
        },
        "Natural Disaster": {
          "count": 15016,
          "production_impact_pct": 37.33,
          "revenue_loss_usd": 3039002.47,
          "full_recovery_days": 90.6,
          "permanent_supplier_change_rate": 0.268
        },
        "Port Congestion": {
          "count": 25118,
          "production_impact_pct": 24.34,
          "revenue_loss_usd": 1963050.99,
          "full_recovery_days": 59.07,
          "permanent_supplier_change_rate": 0.204
        }
      },
      "tests": {
        "production_impact_pct": {
          "test": "one-way ANOVA",
          "p_value": 0.0,
          "significant": true,
          "effect_size_eta_sq": 0.1241,
          "adjusted_eta_sq": 0.0001,
          "adjusted_gap": 0.38
        },
        "revenue_loss_usd": {
          "test": "one-way ANOVA",
          "p_value": 2.63e-188,
          "significant": true,
          "effect_size_eta_sq": 0.0088,
          "adjusted_eta_sq": 0.0,
          "adjusted_gap": 121690.63
        },
... (576 more lines)
```

17. GET /evaluation

Latest offline evaluation results (retrieval, search, classification, DeepEval, RAGAS, judge, models).

Endpoint	HTTP Method
/evaluation	GET

Request format

No request body.

Response format

Key	Type
classification_eval	dict
deepeval_results	dict
demo_scenarios_results	dict
generated_outputs	list
llm_judge_eval	dict
model_benchmark	dict
model_metrics	dict
ragas_results	dict
retrieval_eval	dict
run_notes	dict
search_bias	dict
search_eval	dict

Sample request

```
GET /evaluation
```

Sample response (HTTP 200, lists shortened)


```
{
  "classification_eval": {
    "n_queries": 30,
    "results": {
      "rules": {
        "accuracy": {
          "disruption_type": 1.0,
          "industry": 1.0,
          "supplier_region": 1.0,
          "supplier_tier": 1.0,
          "has_backup_supplier": 1.0
        },
        "severity_MAE": 0.1,
        "severity_within_1": 0.9667
      },
      "llm_agent": {
        "accuracy": {
          "disruption_type": 1.0,
          "industry": 1.0,
          "supplier_region": 1.0,
          "supplier_tier": 1.0,
          "has_backup_supplier": 1.0
        },
        "severity_MAE": 0.4,
        "severity_within_1": 1.0
      }
    }
  },
  "deepeval_results": {
    "framework": "DeepEval",
    "judge_model": "claude-sonnet-5-5 (OpenAI-compatible)",
    "n_cases": 10,
    "summary": {
      "answer_relevancy": {
        "mean": 1.0,
        "n": 10
      },
      "faithfulness": {
        "mean": 1.0,
        "n": 10
      },
      "contextual_precision": {
        "mean": 0.8699,
        "n": 10
      },
      "contextual_recall": {
        "mean": 0.4667,
        "n": 10
      },
      "recommendation_quality": {
        "mean": 0.56,
        "n": 10
      },
      "analysis_correctness": {
        "mean": 0.84,
        "n": 10
      }
    },
    "per_case": [
      {
        "id": "DIS-065446",
        "answer_relevancy": {
          "score": 1.0,
          "passed": true,
          "reason": "The score is 1.00 because the response stayed fully on topic, with no irrelevant statements. It a
    ddressed both parts of the question: what happened in similar semiconductor supply chain cyber incidents, and how
    to respon..."
        },
        "faithfulness": {
          "score": 1.0,
          "passed": true,
          "reason": "The score is 1.00 because there are no contradictions between the actual output and the retrieval
     context. Every claim is fully supported by the provided information. Great job staying faithful to the source mat
    erial!"
```

```
... (1007 more lines)
```

18. POST /feedback

Submit a rating, comment, chosen strategy and optionally the real outcome (used for retraining). At least one signal is required.

Endpoint	HTTP Method
/feedback	POST

Request format

Field	Type	Required	Constraints / allowed values
query	string	no	max len 2000
endpoint	string	no	analyze, recommend, search, classify, other (default analyze)
rating	integer	no	>= 1 <= 5
helpful	boolean	no	
comment	string	no	max len 1000
recommended_strategy	string	no	Alternative Supplier, Production Reroute, Inventory Buffer, Customer Delay, Combined Strategy
chosen_strategy	string	no	Alternative Supplier, Production Reroute, Inventory Buffer, Customer Delay, Combined Strategy
cited_cases	array[string]	no	max items 50
plan_text	string	no	max len 5000
outcome	object (OutcomeReport)	no	disruption_type, industry, supplier_tier, supplier_region, supplier_size, has_backup_supplier, disruption_severity, response_type, production_impact_pct, revenue_loss_usd, response_time_days, partial_recovery_days, full_recovery_days, permanent_supplier_change

Response format

Key	Type
id	int
status	str
strategy_multipliers	dict

Sample request

```
POST /feedback
Content-Type: application/json

{
  "query": "Typhoon at Taiwan chip supplier",
  "endpoint": "analyze",
  "rating": 5,
  "helpful": true,
  "recommended_strategy": "Alternative Supplier",
  "chosen_strategy": "Alternative Supplier",
  "comment": "Clear plan, matched our experience"
}
```

Sample response (HTTP 200, lists shortened)

```
{
  "id": 1,
  "status": "recorded",
  "strategy_multipliers": {
    "Alternative Supplier": 1.04
  }
}
```

19. GET /feedback/stats

Feedback analytics and the learned strategy multipliers.

Endpoint	HTTP Method
/feedback/stats	GET

Request format

No request body.

Response format

Key	Type
total_feedback	int
average_rating	float
helpful_rate	float
by_strategy	dict
rating_trend	dict
strategy_multipliers	dict
outcomes_reported	int
recent	list

Sample request

```
GET /feedback/stats
```

Sample response (HTTP 200, lists shortened)


```
{
  "total_feedback": 1,
  "average_rating": 5.0,
  "helpful_rate": 1.0,
  "by_strategy": {
    "Alternative Supplier": {
      "count": 1,
      "avg_rating": 5.0
    }
  },
  "rating_trend": {
    "2026-10-11": 5.0
  },
  "strategy_multipliers": {
    "Alternative Supplier": 1.04
  },
  "outcomes_reported": 0,
  "recent": [
    {
      "id": 1,
      "created_at": "2026-10-11T15:44:15.058485+00:00",
      "rating": 5,
      "helpful": 1,
      "recommended_strategy": "Alternative Supplier",
      "chosen_strategy": "Alternative Supplier",
      "comment": "Clear plan, matched our experience"
    }
  ]
}
```

20. POST /feedback/retrain

Retrain prediction models including user-reported real outcomes.

Endpoint	HTTP Method
/feedback/retrain	POST

Request format

No request body.

This call retrains every model (several minutes); sample shown for the case with no reported outcomes.

Response format

Key	Type
status	str
reason	str

Sample request

```
POST /feedback/retrain
```

Sample response (HTTP 200, lists shortened)

```
{
  "status": "skipped",
  "reason": "No reported outcomes yet."
}
```